F Prompt Templates

F.1 Sample-Select-Reason Pipeline

In this section, we provide a prompt example for the zero-shot node classification task and detail the prompt template design for the **SSR** pipeline. The pipeline begins with the **Task Description and Initialized Subgraph**, where we define the model's persona as a "graph analysis expert" and provide the raw subgraph data. This stage establishes a comprehensive context, ensuring the model captures both node semantics and explicit topological connections as a foundation for subsequent reasoning.

In the **Sample Phase**, the model is tasked with sampling multiple candidate subgraphs (*e.g.*, 5) while ensuring both their structural

A Prompt Example on Node Classification for SSR pipeline

Task Description and Initialized Subgraph

Task Description:
You are a graph analysis expert. Given a complete subgraph structure, which includes the description of the central node(s), its neighboring nodes, and their connection relationships, please follow the following steps to operate the analysis:

Complete subgraph:
- Central node ID(s): nodex
- Node texts (central node(s) and neighboring nodes): [nodex: node description]
- Connection relationships: [<nodex, nodey>]

Sampling Phase

Step description:
1. **Subgraph Sampling**:
- In order to complete the Node Classification task of the central node(s), you need to first sample **5** subgraphs from the complete graph that are most suitable for the current task from this subgraph.
- These subgraphs must **keep all central node(s)** as mentioned in the complete subgraph, they can include no, partial or all one-hop or two-hop neighbors of the central node(s).
- You must ensure that the sampled subgraphs **actually exist** and do not create fictitious nodes and edges.
- There must be a subgraph that consists **only the central node(s)**.
- There should also be a subgraph, where as many neighboring nodes as possible can **semantically support the central node(s)**. If there is no such subgraph, do not fabricate one.
- You must ensure that the **information differences** between these subgraphs are as significant as possible, so that these subgraphs can represent different semantic and structural meanings. The sizes of the subgraphs should also be as different as possible.

Selection Phase

2. **Subgraph Choosing**:
From the sampled subgraphs, choose one subgraph that is **best suited for the current task** and use it for subsequent reasoning.

Reasoning Phase

3. **Node Classification Reasoning**:
- Repeat and verify the subgraph you have chosen, ensuring that the chosen subgraph you analyze and output **is consistent with** the sampled subgraphs.
- Analyze the chosen subgraph independently. **Only consider the nodes and edges of the current subgraph**. The obtained answer is entirely based on the chosen subgraph and is independent of other sampled subgraphs and the complete subgraph.
- You must **treat the current subgraph as a whole** for reasoning, taking into account the information of all nodes in this subgraph, not just the information of the central node(s).
- Provide the Node Classification answer of the central node(s) obtained through reasoning, the answer must be **selected strictly** from the following Node Classification options. Each option is enclosed within < >, but when you answer, do not include < >.

4. **Node Classification Options**: <Theory>, <Probabilistic Methods>, <Genetic Algorithms>, <Reinforcement Learning>, <Case-Based>, <Neural Networks>, <Rule Learning>.

A Prompt Sample for Subgraph Diversity Score Calculation

Task Description:
You are a graph analysis expert. Given two graph structures, which includes the description of the central node(s), its neighboring nodes, and their connection relationships, please follow the following steps to operate the analysis:

Graph structure 1:
- Central node ID(s): nodex
- Node texts (central node(s) and neighboring nodes): [nodex: node description]
- Connection relationships: [<nodex, nodey>]

Graph structure 2:
- Central node ID(s): nodex
- Node texts (central node(s) and neighboring nodes): [nodex: node description]
- Connection relationships: [<nodex, nodey>]

Step description:
1. Analyze the differences between the two graph structures in terms of the central node(s), neighboring nodes, and connection relationships.
2. Based on the analysis, provide a distance score ranging from 0 to 1 to quantify how different these two graph structures are. A score of 0 indicates that the two graph structures are identical, while a score of 1 indicates that they are completely different.

validity and factual existence. This constraint minimizes the risk of generating "hallucinated" or non-existent subgraphs. By enforcing diverse sampling criteria—ranging from a "target-node-only" structure to "semantically-dense" subgraphs—we guide the model to explore varying scales and semantic contexts. Furthermore, by optimizing for "information difference" between candidates, we encourage the model to consider distinct topological structures and textual attributes, thereby enhancing the overall quality and diversity of the subgraph pool.

In the **Select Phase**, the model is expected to autonomously evaluate the quality of candidate subgraphs and identify the optimal one. To facilitate this, we prescribe only fundamental constraints, granting the model maximum reasoning autonomy. This design allows the model's internal selecting logic to evolve and refine adaptively during the training process, ensuring that the selection criteria align with the overall objective.

In the **Reason Phase**, we impose rigorous constraints by requiring the model to "repeat and verify" the selected subgraphs. This mechanism is designed to maintain logical consistency during long-context generation. By prioritizing independent analysis and holistic modeling, we effectively isolate the decision-making process from the interference of non-selected candidate subgraphs. This ensures that the final classification is derived strictly from the local topology and semantic evidence of the chosen subgraph, yielding the optimal prediction.

F.2 Subgraph Diversity Score Calculation

To capture both topological relationships and textual semantics across heterogeneous datasets, we move beyond traditional GNN-based embeddings. Instead, we employ LLMs to perform direct pairwise distance estimation. We task the LLM with performing a cross-graph structural and semantic analysis to generate a normalized score within the range of $[0, 1]$. This value serves as a universal distance metric to quantify the diversity between two subgraph structures. To further demonstrate the reliability of the LLM-based scoring, we conduct a stability analysis on 100 random samples across 100 independent trials each; the resulting average standard deviation for the Energy score was **0.05147** (relative to the metric's theoretical range of $(-\infty, -1]$), confirming that our LLM-based metric is robust and consistent for quantifying subgraph diversity.

G Training Details

Datasets. To ensure robust model performance across diverse zero-shot graph reasoning tasks, we developed specialized datasets for both SSR-SFT and SSR-RL. Our SSR-SFT training data originates from the publicly available Graph-R1 SFT collection. To adapt these samples to our specific task requirements, we utilized the SSR pipeline to reconstruct the corresponding prompts, which were then processed by DeepSeek-R1 to generate high-quality chain-of-thought reasoning paths and final answers. Following a rigorous quality filtration process, we curated a final SSR-SFT training set comprising 8,265 high-quality instances. Notably, in the Structural Diversity Evaluation, we set a predefined energy threshold of 0.3. For SSR-RL, we implemented a difficulty-aware sampling strategy to enhance the model's exploration capabilities. We first randomly sampled target nodes from the raw graph data and extracted their subgraphs. These subgraphs were transformed into prompts via the SSR pipeline and fed into the previously trained SSR-SFT model for an initial difficulty assessment. Specifically, each instance underwent five independent inference trials. An instance was categorized as "Easy" if the model produced correct results in 4 or 5 trials, "Medium" for 2 or 3 correct results, and "Hard" for 0 or 1 correct results. We sampled across these difficulty tiers with a fixed

ratio of 2:2:1 (Easy:Medium:Hard), yielding a comprehensive SSR-RL training dataset of 10,000 instances. For evaluation, we adopt the datasets utilized in the GOFA paper. All validation instances were processed through the SSR pipeline to maintain consistency in prompt structure with the training phases, ensuring a reliable benchmark.

Baselines. For the baseline results reported in Section 3.2, we directly adopt the performance metrics provided in the Graph-R1 paper. To evaluate the Large Reasoning Models in Section 3.3, we conducted evaluations on the same datasets using a prompt construction method without the SSR pipeline. Given that all experimental settings remain identical across these configurations, the results are directly comparable.

Training Process. We adopt DeepSeek-R1-distilled-Qwen2.5-14B as the base model for all experiments to leverage its superior reasoning capabilities. The training process is divided into two distinct phases. First, in the SSR-SFT phase, we utilize the LlamaFactory framework to fine-tune the model for 3 epochs. We employ a learning rate of 1.0×10^{-5} and a total batch size of 64 to establish a stable foundation for graph-based reasoning.

Subsequently, we transition to the SSR-RL phase using the verl framework. We adopt a sequential training strategy: Authenticity-Reinforced RLVR followed by Denoising-Reinforced RLVR. This phased approach is necessitated by the observation that the model's graph reasoning capabilities following SSR-SFT are insufficient to manage the complexity of subgraph denoising. Introducing the reward r_s prematurely could lead the model to aggressively prune subgraphs, losing vital information before it has learned how to reason on graph-based tasks. By first establishing a baseline of correctness through Authenticity-Reinforced RLVR, the model can then focus on refining its selections in Denoising-Reinforced RLVR.

The overall optimization follows the GRPO loss formulation:

$$\begin{aligned}\mathcal{L}_{GRPO}(\theta) = -\frac{1}{G}\sum_{i=1}^{G}\Bigg[\min\Bigg(&\frac{\pi_\theta(s_i|x)}{\pi_{\theta_{old}}(s_i|x)}\hat{A}_i,\\ &\text{clip}\left(\frac{\pi_\theta(s_i|x)}{\pi_{\theta_{old}}(s_i|x)}, 1-\epsilon, 1+\epsilon\right)\hat{A}_i\Bigg)\\ &-\beta\mathbb{D}_{KL}(\pi_\theta\|\pi_{ref})\Bigg]\end{aligned} \tag{6}$$

where G is the rollout group size and the advantage $\hat{A}_i$ is computed by normalizing the rewards within each group, driving the model to prefer outputs that achieve higher accuracy with lower structural noise.

For RL training, the model is trained for 4 epochs with a more conservative learning rate of 1.0×10^{-6} and a batch size of 64. To prevent the model from being prematurely influenced by the denoising reward—especially when its fundamental graph reasoning capabilities are still maturing—we introduce Denoising-Reinforced RLVR solely during the final 2,048 instances of the RL training process. For the Denoising-Reinforced RLVR, we specifically set the denoising intensity hyperparameter λ to 0.1, balancing the correction of noisy graph structures with the preservation of essential topological information. The entire pipeline, including data synthesis and model training, was executed on a high-performance cluster equipped with 8 NVIDIA H100 GPUs.

H Dataset Descriptions

To verify the cross-dataset adaptability and zero-shot performance of GraphSSR, we conducted experiments on several benchmark datasets spanning five distinct domains: Social Networks, Knowledge Graphs, Citation Networks, E-commerce, and Web Links. Detailed topological statistics for these datasets are summarized in Table 9.

Social Network.

- **Instagram** [17]: This dataset represents a social graph where nodes signify individual users and edges illustrate social ties (e.g., follower-following dynamics). Node-level features are derived from user-provided profile summaries and self-descriptions.

Knowledge Graph.

- **WN18RR** [20]: A subset of the WordNet lexical database, this graph maps relationships between English words. It consists of 40,943 nodes where edges define semantic connections between linguistic entities.
- **FB15K237** [20]: Derived from Freebase, this large-scale knowledge base contains entities (nodes) and their multifaceted relations (edges). Each entity is enriched with textual descriptions based on its name and relational context.

Citation.

- **Arxiv** [11]: A massive citation network of computer science papers. Each node is an academic publication, and edges denote direct citations. The objective is to categorize papers into 40 distinct subfields.
- **Citeseer** [39]: A specialized citation network centered on the computer science domain. It maps the interconnectedness of research papers through their citation links.
- **Pubmed** [9]: A biomedical citation graph sourced from the PubMed database. Nodes represent scientific articles related to diseases, and edges represent citation links.
- **Cora** [36]: A machine learning-centric citation graph. Despite its smaller size, it provides a challenging classification task involving 7 fine-grained categories of research papers.

E-commerce.

- **Children, Computer, Photo, Sports, & Products** [9, 38]: These datasets are product graphs from Amazon. Nodes represent specific products, and edges indicate that products were frequently viewed or purchased in the same session. Textual metadata includes user reviews.

Web Link.

- **WikiCS** [21]: A hyperlink network focused on the computer science branch of Wikipedia. Nodes correspond to individual articles, and directed edges represent the web links connecting them. The full text of each article provides the rich semantic information for each node.

Table 9: Comprehensive statistics of the multi-domain benchmarks. We employ a zero-shot setting, ensuring that datasets marked for testing remain entirely inaccessible during the training phase to rigorously assess model generalizability.

Domain	Dataset	#Nodes	#Edges	#Classes	Split
Social Network	Instagram	11,339	155,349	2	Training
Knowledge Graph	WN18RR	40,943	93,003	11	Training
	FB15K237	**14,541**	**310,116**	**237**	**Test (Zero-shot)**
Citation	Arxiv	169,343	1,166,243	40	Training
	Citeseer	3,186	8,554	6	Training
	Pubmed	19,717	88,648	3	Training
	Cora	**2,708**	**10,556**	**7**	**Test (Zero-shot)**
E-commerce	Children	76,875	1,554,578	24	Training
	Computer	87,229	721,081	10	Training
	Photo	48,362	500,939	12	Training
	Sports	173,055	1,773,500	13	Training
	Products	**54,025**	**144,638**	**47**	**Test (Zero-shot)**
Web Link	**WikiCS**	**11,701**	**216,123**	**10**	**Test (Zero-shot)**

I Computational Cost and Inference Efficiency Analysis

To provide a comprehensive overview of the computational demands of GraphSSR, we systematically evaluate its inference efficiency across all datasets. As presented in Table 10, GraphSSR consumes 6,981.7 tokens with a latency of 208.2s on average, compared to 1,098.4 tokens and 27.6s for Graph-R1.

To investigate the root cause of this variance, we provide a granular step-by-step cost breakdown for GraphSSR in Table 11. The analysis reveals that the computational overhead is heavily concentrated, with over 70% of the resource consumption dedicated to the Sample Phase (57.26%) and answer standardization (14.20%).

Although GraphSSR entails increased inference costs, we argue that this is a necessary and justified trade-off for its capacity to autonomously explore the complex subgraph space and successfully mitigate detrimental structural noise. To further improve practical feasibility, we are also exploring a distillation strategy where GraphSSR serves as a teacher. Specifically, the teacher provides high-quality denoised subgraph examples and accurate reasoning traces to help a student model learn to directly generate an optimized subgraph in a single step. By merging the Sample and Select phases into a unified generative process, we can reduce token consumption and latency while retaining the teacher's denoising expertise. We leave this for future work to balance state-of-the-art performance with computational efficiency.

Table 10: Inference Efficiency Comparison

Metric	Graph-R1	GraphSSR	Diff (%)
Tokens	1,098.4	6,981.7	+535.62%
Latency (s)	27.6	208.2	+654.13%

J Detailed Analysis of Case Study on Cora

To further elucidate the internal mechanisms of the Sample-Select-Reason (SSR) pipeline and the impact of the two-stage RL on subgraph denoising, we present a comprehensive breakdown of the case study referenced in the main text. This example from the Cora dataset illustrates how structural noise in a dense egocentric subgraph can interfere with graph reasoning and how our strategy successfully recovers the precise semantic signal.

J.1 Input Data Characteristics and Semantic Dissonance

The target node (**node 11**) is defined by the text: "*Hierarchical logistic belief networks for linear model selection, using Gibbs sampling for parameter learning*". The ground-truth label is **"Neural Networks"**.

Due to spatial constraints, the "Input Egocentric Subgraph" (*i.e.*, the uniform subgraph used in Graph-R1) exclusively visualizes the target node's first-order neighbors, omitting second-order and higher-order neighbors. These neighboring nodes may provide task-relevant signals, they can also introduce potential noise. Specifically,

- **Relevant Nodes: node 13** (hierarchical mixtures of experts) and **node 17** (representations in neural networks) are all nodes related to the network architecture. They are all directly connected to the target node, forming a "Network" semantic cluster.
- **Noisy Nodes: node 9** (EM algorithm) and **node 14** (EM reinterpreted via free energy maximization) describe the probabilistic method about the EM algorithm. While topically related, their semantic relevance to the target node is not strong enough.

J.2 The Necessity of SSR Pipeline in Task-Relevant Subgraph Denoising

In this section, we provide an extended comparative analysis of the real-world inference results for the case study introduced in

Table 11: Breakdown of GraphSSR Computation Cost

Metric	Sample Phase	Select Phase	Reason Phase	Answer Std.
Tokens	3,262	495	1,129	809
Ratio (%)	57.26%	8.70%	19.83%	14.20%

Section 3.6.1, contrasting GraphSSR with its variant that omits the SSR pipeline. This comparison serves to further justify the necessity of the SSR pipeline and the specialized training regime in isolating task-relevant structural information from inherent graph noise. The detailed variants and results are presented below:

Variant: w/o SSR pipeline. In this variant, the model is forced to process all neighbors simultaneously, including all first-order neighboring nodes and the unlisted second-order neighboring nodes, etc. This results in the model's reasoning focuses on the cumulative occurrence of keywords like "EM algorithm". The signals from the nodes describing "networks" are drowned out by the statistical prevalence of EM-related neighbors. Consequently, the model fails to capture the "Neural Networks" essence of the target node and yields a biased classification of "Probabilistic Methods".

GraphSSR.. The full model GraphSSR, equipped with the SSR pipeline and trained with the two-stage RL, demonstrates a sophisticated filtering strategy. It samples five candidate subgraphs (*Subgraph*_0 to *Subgraph*_4), including a pure subgraph *Subgraph*_2 which contains only node 13 and node 17, and a subgraph *Subgraph*_4 which contains all the first-order neighbors. Due to the existence of the subgraph denoising mechanism, GraphSSR selects *Subgraph*_2 rather than the larger *Subgraph*_4, and "courageously" prunes the noisy nodes describing EM algorithms. By isolating the relationship between "belief networks", "mixtures of experts", and "neural networks", the LLM's reasoning engine recognizes a unified theme: network architecture. This distilled context allows it to bypass the "Probabilistic Methods" category and correctly identify the specific "Neural Networks" category.

J.3 The Synergistic Effect of Two-Stage RL in Balancing Denoising and Reasoning

As illustrated in Section 2.4, we introduce SSR-RL, a novel two-stage reinforcement learning framework, to enhance the model's capacity for both subgraph denoising and complex reasoning. To further demonstrate how this two-stage fusion approach can enhance the ability of subgraph denoising, we conduct a comparative analysis using two distinct model variants: one employing solely Authenticity-Reinforced RLVR and the other restricted to Denoising-Reinforced RLVR.

Variant: solely Authenticity-Reinforced RLVR.. Based on the SSR pipeline, the variant can also sample five candidate subgraphs and include a relatively pure subgraph *Subgraph*_3 in the candidates. However, without the subgraph denoising reward from Denoising-Reinforced RLVR, it incorrectly assumes that more data equals better evidence and chooses *Subgraph*_4, which is the most comprehensive (containing all 4 first-order neighbors). Since *Subgraph*_4 still contains noisy node 9 and node 14, the reasoning is flawed and leads to the incorrect answer "Probabilistic Methods".

Variant: solely Denoising-Reinforced RLVR.. In this variant, there is also a relatively pure subgraph *Subgraph*_1 with only the neighboring node 13 in the candidates. However without Authenticity-Reinforced RLVR, the model has not yet developed a strong graph

The Input Egocentric Subgraph

Complete subgraph:
- **Central node ID(s):** node11
- **Node texts (central node(s) and neighboring nodes):** [
node9: EM algorithm for mixtures of factor analyzers, combining clustering and dimensionality reduction in latent variable models.
node11: Hierarchical logistic belief networks for linear model selection, using Gibbs sampling for parameter learning.
node13: Tree-structured hierarchical mixtures of experts with EM algorithm, validated via robot dynamics simulations.
node14: EM reinterpreted via free energy maximization, enabling faster incremental updates and improved convergence.
node17: Generative models for sparse distributed representations in neural networks, using Bayesian inference and simple learning rules.
]
- **Connection relationships:** [
<node0, node8>, <node1, node12>, <node1, node14>, <node2, node13>, <node3, node13>, <node3, node14>, <node3, node20>, <node4, node13>, <node4, node18>, <node5, node13>, <node6, node7>, <node7, node13>, <node8, node17>, <node9, node11>, <node10, node13>, <node10, node14>, <node10, node18>, <node11, node13>, <node11, node14>, <node11, node17>, <node13, node18>, <node13, node21>, <node15, node18>, <node16, node18>, <node18, node19>
]

reasoning ability at the early stage of training. The premature introduction of the subgraph denoising reward would lead the model to favor smaller, information-sparse subgraphs. Specifically, the model selects *Subgraph*_0—containing only the isolated target node itself—to avoid potential neighbor noise, yet this strategy backfires. Without the structural context of neighboring nodes to clarify its identity, the model is misled by the "Gibbs sampling" description in the target node 11 and also incorrectly yields a "Probabilistic Methods" label.

The empirical results from these two variants demonstrate that Authenticity-Reinforced RLVR and Denoising-Reinforced RLVR serve as indispensable components of GraphSSR. Specifically, the absence of either module results in an imbalance in the adaptive subgraph denoising process—either failing to filter out sufficient noise or inadvertently removing critical structural information. Such an imbalance inevitably leads to suboptimal subgraph quality and a subsequent degradation in overall model performance. By synergistically integrating these two modules, GraphSSR successfully strikes a robust balance between noise reduction and information preservation, ensuring that the generated subgraphs remain both pure and informative, ultimately driving the superior performance of the model across various benchmarks.

The Output of the "w/o SSR Pipeline" Variant

Answer: Probabilistic Methods
Brief_reasoning: Node11 focuses on "hierarchical logistic belief networks" and Gibbs sampling, aligning with probabilistic modeling techniques. Connections to nodes involving EM algorithms (node9, node10, node14), mixture models (node13), and Gibbs sampling (node15, node18) reinforce its probabilistic methodology. While "belief networks" can intersect with neural networks, the emphasis on hierarchical structure and sampling for parameter learning fits better under probabilistic methods. Neural Networks is less precise here, as no direct mention of neural architecture (e.g., layers, activation functions) exists.

The Output of GraphSSR

Subgraph_0
Central_node_ID: node11
Neighboring_node_ID:
Connection_relationship:

Subgraph_1
Central_node_ID: node11
Neighboring_node_ID: node13
Connection_relationship: <node11, node13>

Subgraph_2
Central_node_ID: node11
Neighboring_node_ID: node13,node17
Connection_relationship: <node11, node13>,<node11, node17>

Subgraph_3
Central_node_ID: node11
Neighboring_node_ID: node9,node13,node14,node17
Connection_relationship: <node9, node11>,<node11, node13>,<node11, node14>,<node11, node17>

Subgraph_4
Central_node_ID: node11
Neighboring_node_ID: node13,node4,node18
Connection_relationship: <node11, node13>,<node4, node13>,<node4, node18>,<node13, node18>

Chosen_subgraph_reasoning
Chosen_subgraph: 2
Chosen_subgraph_reason: This subgraph optimally balances relevance and focus: node13 directly supports hierarchical modeling, while node17 explicitly references neural networks, providing strong semantic reinforcement for classification without extraneous nodes.
Central_node_ID: node11
Neighboring_node_ID: node13,node17
Connection_relationship: <node11, node13>,<node11, node17>
Answer: Neural Networks
Brief_reasoning: Node11 describes “hierarchical logistic belief networks” (a neural network architecture). Node13 discusses “hierarchical mixtures of experts” (a neural network model), and node17 explicitly mentions “neural networks” in generative modeling. Collective context confirms neural network focus.

The Output of the "solely Authenticity-Reinforced RLVR" Variant

Subgraph_0
Central_node_ID: node11
Neighboring_node_ID:
Connection_relationship:

Subgraph_1
Central_node_ID: node11
Neighboring_node_ID: node9
Connection_relationship: <node9, node11>

Subgraph_2
Central_node_ID: node11
Neighboring_node_ID: node13,node14
Connection_relationship: <node11, node13>,<node11, node14>

Subgraph_3
Central_node_ID: node11
Neighboring_node_ID: node13,node18
Connection_relationship: <node11, node13>,<node13, node18>

Subgraph_4
Central_node_ID: node11
Neighboring_node_ID: node9,node13,node14,node17
Connection_relationship: <node9, node11>,<node11, node13>,<node11, node14>,<node11, node17>

Chosen_subgraph_reasoning
Chosen_subgraph: 4
Chosen_subgraph_reason: This subgraph contains all direct neighbors of node11, providing maximal semantic context about probabilistic methods (EM, Gibbs sampling) and neural networks, which are central to node11's description.
Central_node_ID: node11
Neighboring_node_ID: node9,node13,node14,node17
Connection_relationship: <node9, node11>,<node11, node13>,<node11, node14>,<node11, node17>
Answer: Probabilistic Methods
Brief_reasoning: The central node (node11) describes hierarchical logistic belief networks using Gibbs sampling. Neighbors node9, node13, and node14 focus on EM algorithms for probabilistic models, while node17 discusses generative models with Bayesian inference. Collectively, this subgraph emphasizes probabilistic modeling techniques, aligning with the "Probabilistic Methods" classification.

The Output of the "solely Denoising-Reinforced RLVR" Variant

Subgraph_0
Central_node_ID: node11
Neighboring_node_ID:
Connection_relationship:

Subgraph_1
Central_node_ID: node11
Neighboring_node_ID: node13
Connection_relationship: <node11, node13>

Subgraph_2
Central_node_ID: node11
Neighboring_node_ID: node13,node14
Connection_relationship: <node11, node13>,<node11, node14>

Subgraph_3
Central_node_ID: node11
Neighboring_node_ID: node17,node8
Connection_relationship: <node11, node17>,<node8, node17>

Subgraph_4
Central_node_ID: node11
Neighboring_node_ID: node9,node13,node14,node17,node18
Connection_relationship: <node9, node11>,<node11, node13>,<node11, node14>,<node11, node17>,<node13, node18>

Chosen_subgraph_reasoning
Chosen_subgraph: 0
Chosen_subgraph_reason: Subgraph_0 is optimal as it isolates the central node's core semantics ("belief networks" and "Gibbs sampling"), avoiding potential neighbor-induced biases (e.g., neural network terminology in other subgraphs). The central node's text alone provides unambiguous evidence for classification.
Central_node_ID: node11
Neighboring_node_ID:
Connection_relationship:
Answer: Probabilistic Methods
Brief_reasoning: The central node (node11) explicitly describes hierarchical logistic belief networks—a probabilistic graphical model, and uses Gibbs sampling—a core probabilistic inference technique. This directly aligns with "Probabilistic Methods" without requiring neighbor context.